

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

DATE	12 July 2026
TEAM ID	
PROJECT NAME	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
MAXIMUM MARKS	4 Marks

FUNCTIONAL REQUIREMENTS

FR NO.	FUNCTIONAL REQUIREMENT (EPIC)	SUB REQUIREMENT (STORY / SUB-TASK)
FR-1	Data Collection & Import	Import electricity consumption dataset into Tableau
		Support CSV/Excel data format
FR-2	Data Processing	Clean and prepare dataset for visualization (remove null values, formatting)
		Create calculated fields for year, month and regional metrics
		Categorize states into regions (WR, NR, SR, ER, NER)
FR-3	Visualization Creation	Create yearly electricity consumption trend chart
		Create region-wise consumption pie chart
		Create Top N and Bottom N states bar charts
		Create month-wise consumption charts
		Create India state consumption maps
FR-4	Dashboard Development	Build interactive dashboards combining multiple visualizations
		Enable filtering by year, region, and state
		Add titles, legends and labels for clarity
		Provide comparative view of 2019 vs 2020 data

NON-FUNCTIONAL REQUIREMENTS

NFR NO.	NON-FUNCTIONAL REQUIREMENT	DESCRIPTION
NFR-1	Usability	Dashboard should be easy to understand and interactive for non-technical users. Visualizations must be clear and self-explanatory.
NFR-2	Security	Dataset used must be public/non-confidential and protected from unauthorized modification.
NFR-3	Reliability	Dashboard should display accurate and consistent results based on the dataset.
NFR-4	Performance	Dashboard should load and respond to filters quickly without lag.
NFR-5	Availability	Dashboard should be accessible anytime through Tableau Public.
NFR-6	Scalability	Solution should allow addition of future years and more states without redesigning the dashboard.